# Application of AI Models in predicting outcomes of Lung Pathologies

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## Abstract

Lung pathologies encompass a diverse range of respiratory conditions, including asthma, chronic obstructive pulmonary disease (COPD), pneumonia, tuberculosis, and lung cancer, each exhibiting distinct pathophysiological characteristics. The prevalence of pulmonary disorders, particularly highlighted by the nearly 3.5 million cases reported globally in 2020, underscores their significant impact on public health, as these diseases rank as the third leading cause of mortality worldwide. Mechanical ventilation can complicate the clinical course for many patients, particularly due to the risk of ventilator-associated pneumonia (VAP). As diagnostic accuracy becomes increasingly critical, artificial intelligence (AI) and machine learning (ML) models are being integrated into clinical practice, significantly enhancing the diagnostic and prognostic capabilities for lung diseases.

This narrative review aims to explore the application of AI and ML models in predicting and diagnosing outcomes in lung pathologies. The investigation reveals that AI models effectively diagnose various lung conditions, with deep learning models such as PulmoNet achieving an impressive accuracy of up to 99.40% in identifying lung disorders from radiographic images. Additionally, the review highlights the potential of these technologies in predicting clinical outcomes; for instance, studies employing XGBoost and Support Vector Machines (SVM) yielded accuracies of 90.31% and 96.00%, respectively, in the context of chronic obstructive pulmonary disease (COPD). Moreover, the evaluation of machine learning models in asthma classification demonstrated varied performance, with Random Forest achieving an accuracy of 92.10% alongside high precision and sensitivity metrics.

However, the integration of AI and ML into clinical settings is not without challenges. Issues such as variability in dataset characteristics, the need for large and diverse datasets, and the importance of interpretability remain significant barriers to widespread adoption. This review provides a comprehensive overview of the current state of AI and ML in lung pathology, emphasizing the urgent need for advanced diagnostic tools and predictive models to enhance patient outcomes in this critical area of healthcare.

# Introduction

Lung pathologies comprise a broad spectrum of respiratory conditions, ranging from acquired diseases to congenital defects. Key conditions include asthma, chronic obstructive pulmonary disease (COPD), pneumonia, tuberculosis and lung cancer, each with distinct pathophysiological features[1-3]. With nearly 3.5 million pulmonary disorders, including Covid-19-related deaths, reported globally in 2020 alone, pulmonary diseases rank third in terms of causes of death globally [4]. A major risk for individuals receiving mechanical ventilation is ventilator-associated pneumonia (VAP)[5]. The lungs are one of the target organs for connective tissue diseases (CTD) that sustain damage most frequently (20%–95% of patients)[6]. Based on underlying diseases, pathogen exposure, and individual health considerations, lung disorders vary greatly in severity. The diversity of lung pathologies complicates accurate diagnosis[7-9]. Accuracy and efficiency in lung pathology diagnosis and treatment have significantly improved with the use of machine learning (ML) models [10]. Clinical decision-making is improved by these models by utilizing a variety of data types, such as auscultation signals and imaging data [11, 12].These statistics underscore the urgent need for advanced diagnostic tools and predictive models to improve outcomes in patients with lung pathologies [13, 14].

Artificial Intelligence (AI) and machine learning (ML) are increasingly recognized as transformative technologies in this field, offering significant potential to enhance the diagnosis, prognosis, and treatment of lung diseases [15, 16]. Machine learning, a branch of artificial intelligence, allows computers to learn and get better on their own without explicit programming by finding patterns in data, predicting future events, and making decisions [17, 18]. With excellent accuracy rates of up to 99.40%, the PulmoNet model uses deep learning techniques to identify lung disorders from radiographic images, such as pneumonia and COVID-19[4, 19]. In numerous trials, LungNet, a deep learning architecture, outperforms conventional techniques in diagnosing respiratory disorders by using multi-dimensional representations of lung sounds[20]. Radiographic pictures and lung auscultation sounds are well-analyzed by Convolutional Neural Networks (CNN) and hybrid models such as CNN-LSTM. The CNN-LSTM model classified respiratory diseases with up to 100.00%% accuracy[21-23]. Additionally, clinical and genomic data are incorporated into ML models to improve the detection of biomarkers and the course of disease[24].

These models help in clinical decision-making, improve diagnostic accuracy, and forecast clinical outcomes. Certain models such as Support Vector Machines (SVM) and Random Forests (RF) have been successful in identifying subtypes of non-small cell lung cancer, with AUC values of 87.60% and 86.30% respectively[23, 25].

This narrative review aims to explore the application of AI and machine learning models in predicting and diagnosing outcomes in lung pathologies. The specific objectives of this review are:

1. To investigate the role of AI models in the diagnosis of various lung pathologies, including asthma, COPD, pneumonia, tuberculosis and lung cancer.
2. To evaluate the predictive capabilities of these models in forecasting clinical outcomes and treatment responses.
3. To analyze the challenges and limitations associated with integrating machine learning models into clinical practice for lung disease management.

# Methods

This review on the application of AI models in predicting outcomes of lung pathologies is organized into four primary stages: a comprehensive literature search, study selection, data extraction, and the synthesis of findings.

The literature search was systematically conducted across numerous academic databases, including Respiratory Medicine, the European Respiratory Journal, the Journal of Thoracic Disease, the Journal of Pulmonary Medicine, PubMed, IEEE Xplore, and Google Scholar. A broad range of keywords was used to capture relevant studies on the application of machine learning in respiratory diseases. These terms included "AI models in lung disease," "machine learning for respiratory pathology," "supervised learning for lung conditions," "deep learning for lung cancer detection," "AI in COPD prediction," "predictive modeling for respiratory diseases," "ensemble learning in pulmonary care," "hybrid models for respiratory conditions," "Random Forest in lung disease," "Support Vector Machines for pulmonary pathology," "CNN for lung diagnostics," and "artificial intelligence in pulmonary disease management." Additionally, relevant references from key articles were reviewed to identify additional studies.

Next, the process of study selection was carried out, focusing on articles that specifically addressed machine learning applications in diagnosing, predicting, and managing various lung diseases, including COPD, pneumonia, lung cancer, and interstitial lung diseases. Studies were prioritized based on their comprehensive analysis of the performance and utility of machine learning models across a range of lung pathologies. Papers that explored both the benefits and challenges of AI integration in clinical settings were particularly emphasized. To ensure relevance and accuracy, only peer-reviewed publications and systematic reviews from the past ten years were included. Research from before 2010, non-peer-reviewed material, and studies not available in English were excluded unless reliable translations were available.

For data extraction, a standardized process was employed to systematically capture key information from each selected study. This included the type of machine learning model utilized—whether supervised, deep learning, ensemble, or hybrid—and the specific algorithms implemented, such as Random Forest, Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), and Support Vector Machines (SVM). This structured approach facilitated a detailed comparison of the performance of various machine learning models applied to different lung pathologies.

In the synthesis of findings, the review focused on evaluating and comparing the effectiveness of machine learning models in diagnosing and predicting outcomes in lung diseases. Supervised learning models were assessed based on their accuracy in identifying respiratory conditions using labeled data. The diagnostic capabilities of models such as Random Forest, K-Nearest Neighbors (KNN), SVM, and Logistic Regression were examined for various lung diseases, including lung cancer, pneumonia, and COPD. Additionally, deep learning models like CNNs and hybrid architectures (such as CNN-LSTM) were evaluated for their ability to analyze complex data, including medical images and lung sounds. The goal of this analysis was to highlight how these models contribute to greater diagnostic accuracy, minimize errors, and improve patient care in respiratory medicine.

This methodology allowed for a detailed and systematic review of AI models used in lung pathology prediction, focusing on their effectiveness and potential applications in clinical practice.

## Results

The results section of this review provides a thorough analysis of various machine learning models used in the diagnosis and treatment of various lung pathologies including asthma, chronic obstructive pulmonary disease (COPD), pneumonia, lung cancer and tuberculosis. It evaluates the effectiveness, strengths, and limitations of models such as logistic regression (LR), random forest (RF), support vector machine (SVM), k-nearest neighbors (KNN), decision tree (DT), artificial neural networks (ANN), convolutional neural networks (CNN), Long Short-Term Memory (LSTM), XGBoost, and hybrid models. By synthesizing data from multiple studies, this section highlights how these models enhance diagnostic accuracy, improve treatment strategies, and positively impact patient outcomes in pulmonary medicine. The review aims to provide a comprehensive overview of the role these machine learning models play in advancing lung disease diagnosis and management, as well as their potential for future clinical applications.

The following sections present the reviewed studies, outlining their strengths, limitations, and possible integration into clinical practice.

### Asthma

Asthma is a long-term inflammatory condition of the airways, marked by recurring episodes of reversible airflow blockage due to bronchial hyper-reactivity[26, 27]. Excess of mast cells, activated T helper lymphocytes and eosinophils release mediators that trigger bronchoconstriction, mucus secretion, and remodeling that contribute to the persistent inflammation of the airways[28]. According to the latest report of the World Health Organization (WHO), the number of asthmatics in the world is 300 million, which is estimated to increase to 400 million by 2025[29]. In 2016, the Centers for Disease Control stated that 8.3% of children in the United States suffer from asthma[30, 31]. A common symptom of asthma, wheezing is not always indicative of asthma because it can also be caused by other illnesses like bronchiolitis. Predicting which children with persistent wheeze would eventually develop asthma is the main challenge. Using machine learning models to make accurate predictions could greatly improve management and preventive actions[32].

*Table 1 : Overview of Machine Learning Models in Asthma*

| Case Study | Author(s) | Model Type | Dataset Size | Accuracy | Precision | Specificity | Sensitivity |
|---|---|---|---|---|---|---|---|
| **1** | Luo, et al. [33] | XGBoost | 334,564 patients | 90.31% | 22.65% | 91.93% | 53.69% |
| **2** | Joo, et al. [34] | XGBoost | 4,235 patients | 87.10% | N/A | 97.90% | 82.50% |
| **3** | Xie and Xu [35] | XGBoost | 9,716 youth subjects | 87.55% | 33.58% | N/A | 14.47% |
| **4** | Tong, et al. [36] | XGBoost | 31724 datasets from UWM | 88.20% | 85.00% | N/A | 88.00% |

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|---|---|---|---|---|---|---|---|
| **5** | Inselman, et al. [37] | GBoost | 3057 patients | 74.00% | N/A | N/A | N/A |
| **6** | Joo, et al. [34] | LGBM | 4,235 patients | 85.88% | N/A | 78.95% | 100.00% |
| **7** | Joo, et al. [34] | RF | 4,235 patients | 83.53% | N/A | 59.87% | 100.00% |
| **8** | Ullah, et al. [38] | RF | 150 asthmatic patients | 92.10% | 92.40% | 76.90% | 97.30% |
| **9** | Xie and Xu [35] | RF | 9,716 youth subjects | 89.06% | 47.06% | N/A | 2.52% |
| **10** | Inselman, et al. [37] | RF | 3057 patients | 75.00% | N/A | N/A | N/A |
| **11** | Goto, et al. [39] | RF | NHAMCS dataset. (3896) | N/A | 53.00% | 76.00% | 75.00% |
| **12** | Sills, et al. [40] | RF | Triage dataset | 77.70% | N/A | N/A | 63.50% |
| **13** | Tong, et al. [36] | RF | 31724 datasets from UWM | 87.87% | 86.00% | N/A | 87.00% |
| **14** | Joo, et al. [34] | DT | 4,235 patients | 81.18% | N/A | 71.93% | 100.00% |
| **15** | Farion, et al. [41] | DT | Extracted from ED charts | 83.00% | N/A | 71.00% | 84.00% |
| **16** | Ullah, et al. [38] | SVM | 150 asthmatic patients | 94.10% | 94.20% | 82.70% | 98.00% |
| **17** | Tong, et al. [36] | SVM | 31724 datasets from UWM | 84.90% | 81.00% | N/A | 82.00% |
| **18** | Topaloglu, et al. [42] | SVM | 1489 patients | 99.73% | 99.74% | 99.72% | 99.74% |
| **19** | Finkelstein and Jeong [31] | SVM | 7001 records | 95.20% | N/A | 98.20% | 45.50% |
| **20** | Ullah, et al. [38] | ANN | 150 asthmatic patients | 92.10% | 94.70% | 84.60% | 94.70% |
| **21** | Xie and Xu [35] | ANN | 9,716 youth subjects | 88.23% | 54.49% | N/A | 11.32% |
| **22** | Goto, et al. [39] | DNN | NHAMCS dataset. (3896) | N/A | 53.00% | 77.00% | 73.00% |
| **23** | Darsha Jayamini, et al. [43] | Naïve Bayes | 860 studies | 70.70% | 85.00% | 73.00% | 70.00% |
| **24** | Tong, et al. [36] | Naïve Bayes | 31724 datasets from UWM | 46.04% | 71.00% | N/A | 46.00% |
| **25** | Finkelstein and Jeong [31] | Naive Bayesian | 7001 records | 82.30% | N/A | 83.50% | 63.90% |
| **26** | Xie and Xu [35] | LR | 9,716 youth subjects | 88.78% | 40.82% | N/A | 6.29% |
| **27** | Sills, et al. [40] | LR | Triage dataset | 73.10% | N/A | N/A | 56.4% |
| **28** | Goto, et al. [39] | LR | NHAMCS dataset. (3896) | N/A | 53.00% | 77.00% | 73.00% |
| **29** | Inselman, et al. [37] | GLMnet | 3057 patients | 72.00% | N/A | N/A | N/A |
| **30.** | Goto, et al. [39] | GBDT | NHAMCS dataset. (3896) | N/A | 53.00% | 76.00% | 73.00% |

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| **31** | Joumaa, et al. [44] | Boosting models | LPD dataset | N/A | 71.00% | 63.00% | 83.00% |
| **32.** | Joumaa, et al. [44] | RNN | LPD dataset | N/A | 71.00% | 62.00% | 83.00% |
| **33.** | Tong, et al. [36] | C4.5 | 31724 datasets from UWM | 87.37% | 84.00% | N/A | 87.00% |
| **34.** | Tong, et al. [36] | KNN | 31724 datasets from UWM | 59.62% | 60.00% | N/A | 60.00% |
| **35.** | Topaloglu, et al. [42] | KNN | 1489 patients | 99.19% | 99.48% | 99.45% | 98.96% |

The case studies below are classification tasks for models used to classify patients' categories or predict disease presence in asthma.

Luo, et al. [33] employed XGBoost and achieved an accuracy of 90.31% but a low precision of 22.65%, while. Joo, et al. [34] also used XGBoost with 4,235 patients, reporting an accuracy of 87.10%, high specificity (97.90%), and sensitivity (82.50%). Xie and Xu [35] used XGBoost, attaining an accuracy of 87.55% and a precision of 33.58%, but a low sensitivity of 14.47%. Tong, et al. [36] investigated XGBoost, achieving an accuracy of 88.20% and a high precision of 85%, along with a sensitivity of 88%. Random Forest (RF) was widely used across studies. Ullah, et al. [38] achieved an accuracy of 92.10% with high precision (92.40%) and sensitivity (97.30%). Xie and Xu [35] applied RF on their dataset, achieving 89.06% accuracy but a very low sensitivity of 2.52%. Goto, et al. [39] used RF and attained a precision of 53% and sensitivity of 75%. Tong, et al. [36] reported an accuracy of 87.87% with RF exhibiting high precision (86%). Sills, et al. [40] and Inselman, et al. [37] achieved an accuracies of 77.70% and 75.00% respectively.

Ullah, et al. [38] achieved 94.10% accuracy, high precision (94.20%), and sensitivity (98.00%) using SVM. Topaloglu, et al. [42] reported an impressive accuracy of 99.73% with SVM, alongside near-perfect precision and sensitivity. Finkelstein and Jeong [31] obtained 95.20% accuracy with SVM but faced a low sensitivity of 45.50% while  Tong, et al. [36] had an accuracy of 87.875%. Artificial Neural Networks (ANN) showed promise as well. Ullah, et al. [38] achieved 92.10% accuracy and high sensitivity (94.70%). Xie and Xu [35] obtained an accuracy of 88.23% but a significantly lower sensitivity of 11.32%. Naïve Bayes was utilized by Darsha Jayamini, et al. [43], achieving 70.70% accuracy and 85% precision. Tong, et al. [36] reported a lower accuracy of 46.04%, but with a precision of 71%. Finkelstein and Jeong [31] achieved 82.30% accuracy with Naïve Bayes, alongside 83.50% specificity. Decision Trees (DT) also featured prominently, with Joo et al. [34] achieving 81.18% accuracy and perfect sensitivity (100%) but lower specificity (71.93%). Farion, et al. [41] reported 83% accuracy with DT while Joo, et al. [34] achieved an accuracy of 81.18%. Xie and Xu [35] applied logistic regression, achieving an accuracy of 88.78%, but struggled with a precision of only 40.82% and a low sensitivity of 6.29%. Similarly, Sills, et al. [40] reported a modest accuracy of 73.1% with a sensitivity of 56.4% in their triage dataset.

Further advancements were noted in studies utilizing boosting models and recurrent neural networks (RNN), with Joumaa, et al. [44] achieving comparable precision and specificity metrics of 71.00% and 63.00% respectively. Topaloglu, et al. [42] and Tong, et al. [36] used KNN with accuracies of 99.19% and 59.62% respectively.

## Chronic Obstructive Pulmonary Disease (COPD)

Chronic obstructive pulmonary disease (COPD) affects roughly 250 million people globally and the third leading cause of mortality globally, carrying a substantial economic impact[45, 46]. Enrolling patients who are at high risk in a care management program for preventative treatment is a commonly used strategy to reduce severe COPD exacerbations. A predictive model can be used to prospectively identify patients who are at high risk[47]. Every year, there are approximately 200 million instances of viral community-acquired pneumonia—100 million in children and 100 million in adults[48, 49]. There are 12 attacks for every 1,000 people in a year[50]. The primary cause of the global child death, according to the World Health Organization (WHO), is pneumonia. Roughly 1.2 million children under five have perished as a result of it[51]. Determining the treatment site and identifying patients at high risk of pneumonia require early and accurate assessment of the disease's severity[52].

*Table 2 : Overview of Machine Learning Models in COPD*

| Case Study | Author(s) | Model Type | Dataset Size | Accuracy | Precision | Specificity | Sensitivity |
|---|---|---|---|---|---|---|---|
| **1** | Zhu, et al. [53] | CNN | 2,983 records | 78.20% | 88.20% | N/A | N/A |
| **2** | Amaral, et al. [54] | KNN | 4,235 patients | 97.00% | N/A | 99.00% | 96.00% |
| **3** | Hussain, et al. [55] | KNN | 2900 patients | 86.36% | 88.41% | N/A | N/A |
| **4** | Wang, et al. [56] | KNN | 6648 patients | 74.30% | N/A | N/A | 71.20% |
| **5** | Amaral, et al. [54] | DT | 4,235 patients | 90.00% | N/A | 90.00% | 91.00% |
| **6** | Maldonado-Franco, et al. [57] | DT | 695 patients | 88.60% | 84.60% | 96.20% | 64.70% |
| **7** | Wu, et al. [58] | DT | 5600 datasets | 79.20% | N/A | N/A | 71.20% |
| **8** | Amaral, et al. [54] | SVM | 4,235 patients | 96.00% | N/A | 94.00% | 97.00% |
| **9** | Maldonado-Franco, et al. [57] | SVM | 695 patients | 85.70% | 70.60% | 90.60% | 70.60% |
| **10** | Hussain, et al. [55] | SVM | 2900 patients | 88.18% | 86.91% | N/A | N/A |
| **11.** | Kor, et al. [59] | SVM | 509 patients | 73.53% | 57.14% | 69.12% | 82.35% |
| **12** | Maldonado-Franco, et al. [57] | ANN | 695 patients | 92.90% | 83.30% | 94.30% | 88.20% |
| **13** | Hussain, et al. [55] | XGBoost | 2900 patients | 88.48% | 86.48% | N/A | N/A |
| **14** | Muro, et al. [60] | XGBoost | 20,265 individuals | 91.70% | N/A | 96.00% | 84.50% |
| **15** | Kor, et al. [59] | XGBoost | 509 patients | 77.45% | 68.97% | 86.76% | 58.82% |
| **16** | Tong, et al. [61] | XGBoost | 82,888 patients | 90.20% | 10.45% | 90.91% | 70.20% |

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| **17** | Wang, et al. [56] | XGBoost | 6648 patients | 83.40% | N/A | 98.30% | 5.10% |
| **18** | Tong, et al. [61] | XGBoost | 334,564 patients | 85.9% | 22.65% | 91.93% | 53.69% |
| **19** | Kor, et al. [59] | Gradient Boost | 509 patients | 78.43% | 64.29% | 77.94% | 79.41% |
| **20** | Muro, et al. [60] | LR | 20,265 individuals | 88.40% | N/A | 94.60% | 87.40% |
| **21** | Wang, et al. [56] | LR | 6648 patients | 84.10% | N/A | 99.3% | 3.9% |
| **22** | Wang, et al. [62] | LR | 303 electronic medical records | N/A | 78.00% | 79.00% | 83.00% |
| **23** | Goto, et al. [63] | LR | 44,929 medical records | N/A | 10.00% | 51.00% | 66.00% |
| **24** | Goto, et al. [39] | LR | NHAMCS dataset. (3896) | N/A | 53% | 77% | 73% |
| **25** | Goto, et al. [39] | RF | 5807 participants | 90.50% | N/A | 98.60% | 8.20% |
| **26** | Wang, et al. [62] | RF | 303 electronic medical records | N/A | 72.00% | 71.00% | 83.00% |
| **27** | Wu, et al. [58] | RF | 5600 datasets | 91.40% | N/A | NA/A | 87.7% |
| **28** | Goto, et al. [39] | RF | NHAMCS dataset. (3896) | N/A | 53% | 76% | 75% |
| **29** | Kor, et al. [59] | RF | 509 patients | 77.45% | 66.67% | 83.82% | 64.71% |
| **30** | Wang, et al. [62] | Naïve Baye | 303 electronic medical records | N/A | 71.00% | 65.00% | 85.00% |
| **31** | Wu, et al. [58] | LDA | 5600 datasets | 82.90% | N/A | N/A | 78.10% |
| **32** | Wu, et al. [58] | AdaBoost | 5600 datasets | 88.60% | N/A | N/A | 82.20% |
| **33** | Wu, et al. [58] | DNN | 5600 datasets | 92.10% | N/A | N/A | 90.40% |
| **34** | Goto, et al. [39] | DNN | NHAMCS dataset. (3896) | N/A | 53.00% | 77.00% | 73.00% |
| **35** | Goto, et al. [39] | GBDT | NHAMCS dataset. (3896) | N/A | 53.00% | 76.00% | 73.00% |
| **36** | Joumaa, et al. [44] | Boosting models | LPD dataset | N/A | 61.00% | 90.00% | 65.00% |
| **37** | Joumaa, et al. [44] | RNN | LPD dataset | N/A | 66.00% | 91.00% | 65.00% |

The case studies below are classification tasks for models used to classify patients' categories or predict disease presence in COPD.

Convolutional Neural Networks (CNN) are known for their effectiveness in image recognition, and Zhu, et al. [53] applied this model, achieving an accuracy of 78.20% and an impressive precision of 88.20%. Similarly, K-Nearest Neighbors (KNN) was tested across several studies with varying results. Amaral, et al. [54] recorded a high accuracy of 97.00%, coupled with a specificity of 99.00% and sensitivity of

96.00%, demonstrating KNN's robustness. In contrast, Hussain, et al. [55] and Wang, et al. [56] reported lower accuracy values of 86.36% and 74.30%, respectively, highlighting KNN's dependence on dataset characteristics. Amaral, et al. [54] achieved an accuracy of 90.00% with DT, which was supported by high specificity (90.00%) and sensitivity (91.00%). However, Wu, et al. [58] and Maldonado-Franco, et al. [57] reported lower accuracies of 79.20% and 88.60%, respectively. Maldonado-Franco, et al. [57] also observed a notable disparity between precision (84.60%) and sensitivity (64.70%), highlighting DT's inconsistency in maintaining high sensitivity. Support Vector Machines (SVM) have shown robust performance in various studies. Amaral, et al. [54] achieved an accuracy of 96.00% with high specificity (94.00%) and sensitivity (97.00%). However, Kor, et al. [59] observed lower accuracy (73.53%) and precision (57.14%) in their study, showcasing SVM's variability with smaller datasets. Hussain, et al. [55] and Maldonado-Franco, et al. [57] also reported accuracies of 88.18% and 85.70% respectively.

Boosting techniques such as XGBoost and Gradient Boosted Decision Trees (GBDT) have gained significant traction in medical diagnostics. Muro, et al. [60] demonstrated the strength of XGBoost, achieving an accuracy of 91.70%, with a sensitivity of 84.50% and specificity of 96.00%. On the other hand, studies by Tong, et al. [61] and Kor, et al. [59] found XGBoost's performance less consistent, reporting accuracies of 90.20% and 77.45%, respectively. Tong, et al. [61] reported a significantly low precision of 10.45%, indicating that while XGBoost models generally perform well, precision can fluctuate depending on the dataset. Gradient Boost also showed mixed results, with Kor, et al. [59] achieving an accuracy of 78.43% and balanced sensitivity (79.41%) and specificity (77.94%).

Logistic Regression (LR), a widely-used algorithm in medical research, was evaluated across multiple studies. Muro, et al. [60] reported an accuracy of 88.40%, with a specificity of 94.60% and sensitivity of 87.40%. However, Goto, et al. [63] found LR's precision to be low at 10.00%, despite higher specificity (51.00%) and sensitivity (66.00%). Random Forest (RF) models also showed varied performance, with Goto, et al. [63] achieving an accuracy of 90.50%, but Kor et al. and Wu et al. observing lower accuracies of 77.45% and 91.40%, respectively. Also, Goto, et al. [39], reported an accuracy of 90.50%

Advanced techniques such as deep neural networks (DNNs) and recurrent neural networks (RNNs) were also evaluated. Wu, et al. [58] demonstrated DNN's strength with an accuracy of 92.10% and a sensitivity of 90.40%. RNNs, often used for time-series data, were applied by Joumaa, et al. [44], who reported modest performance with an accuracy of 66.00% and precision of 91.00%.

## Lung Cancer

Lung cancer is the leading cause of global cancer incidence with an estimated 2 million diagnoses and 1.8 million deaths per year[64-66]. Lung cancer is at the top of the list partly because it is frequently discovered only when the disease is at the advanced stage[65, 67]. By utilizing machine learning, robust algorithms can identify lung cancer at its nascent stage. A model of this kind might be very helpful to doctors in helping them decide whether a patient needs an intensive diagnosis or a standard one during the diagnostic procedure[68].

*Table 3 : Overview of Machine Learning Models in Lung Cancer*

| Case Study | Author(s) | Model Type | Dataset Size | Accuracy | Precision | Specificity | Sensitivity |
|---|---|---|---|---|---|---|---|
| **1** | Al-Sheikh, et al. [69] | Pre-trained VGG16Net | 904 X-ray images. | 98.20% | 98.15% | 98.30% | 98.30% |
| **2** | Al-Sheikh, et al. [69] | Pre-trained AlexNet | 904 X-ray images. | 98.20% | 98.20% | 98.30% | 98.20% |
| **3** | Pathan, et al. [70] | SVM | Data world. 1000 images of different features | 96.00% | 98.00% | N/A | 100.00% |
| **4** | Rodrigues, et al. [71] | SVM | LIDC/IDRI | 96.70 | 98.20% | 95.20 | 96.75% |
| **5** | Orozco, et al. [72] | SVM | NBIA-ELCAP (113) | N/A | N/A | 52.17% | 96.15% |
| **6** | Khehrah, et al. [73] | SVM | LIDC-IDRI | 92.00% | 83.30% | 91.20% | 93.70% |
| **7** | Halder, et al. [74] | SVM | LIDC-IDRI | 88.20% | N/A | 86.9 | 86.9 |
| **8** | Pathan, et al. [70] | KNN | Data world. 1000 images of different features | 99.00% | 100.00% | N/A | 100.00% |
| **9** | Rodrigues, et al. [71] | KNN | LIDC/IDRI | 95.30% | 96.40% | 94.20% | 95.30% |
| **10** | Sharma, et al. [75] | SVM + k-NN | SPIE-AAPM Lung CT Challenge | 93.90% | N/A | N/A | 94.50% |
| **11** | Krewer, et al. [76] | DT, KNN and SVM | LIDC-IDRI (33) | 90.91% | N/A | 94.74% | 85.71% |
| **12** | Pathan, et al. [70] | DT | Data world. 1000 images of different features | 100.00% | 100.00% | N/A | 100.00% |
| **13** | Pathan, et al. [70] | RF | Data world. 1000 images of different features | 100.00% | 100.00% | N/A | 100.00% |
| **14** | Rodrigues, et al. [71] | MLP | LIDC/IDRI | 95.40% | 98.20% | 92.60% | 95.52% |
| **15** | Masud, et al. [77] | CNN model | LC25000 dataset. 1250 images were used. | 96.33% | 96.39% | N/A | 96.37% |
| **16** | Song, et al. [78] | CNN model | (LIDC) dataset | 84.15% | N/A | 84.32% | 83.965 |
| **17** | Wang, et al. [79] | CNN | LIDI-IDRI database | 82.30% | N/A | 83.80% | 79.40% |
| **18** | El-Regaily, et al. [80] | Multi-view CNN | LIDC-IDRI | 91.00% | N/A | 87.30% | 96.00% |
| **19** | Liao, et al. [81] | 3D CNN and a leaky noisy-OR gate | LUNA16 data set and training set of DSB 2017 | 81.40% | N/A | N/A | N/A |

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|---|---|---|---|---|---|---|---|
| **20** | Crasta, et al. [82] | 3D-ResNet Model | LUNA16 dataset. 1018 thoracic CT scans. | 99.20% | N/A | 99.6% | 98.8% |
| **21** | Wu, et al. [83] | Deep residual network | LIDC-IDRI | 98.20% | 98.50% | 98.30% | 97.70% |
| **22** | Silva, et al. [84] | Deep residual network | LIDC-IDRI | 93.10% | 83.80% | N/A | 81.70% |
| **23** | Zhou, et al. [85] | VNet | LUNA16 dataset | 84.80% | N/A | 99.70% | 96.70% |
| **24** | Bharti, et al. [86] | Vnet and 3D Resnet | LUNA16 dataset | 95.00% | 98.70% | N/A | 98.20% |
| **25** | Ye, et al. [87] | VNet model and SVM classifier | Luna16 dataset | 66.70% | 70.80% | N/A | 93.7% |
| **26** | Dodia, et al. [88] | Regularized VNet, NCNet | LUNA16 dataset | 98.20% | N/A | N/A | 98.30% |
| **27** | Bansal, et al. [89] | Deep3DSCan | LUNA16 dataset | 88.30% | 90.00% | 89.70% | 87.10% |
| **28** | Jiang, et al. [90] | Ensembling 3D-Dual Path Networks | LUNA16 dataset | 90.24% | N/A | N/A | 92.04% |
| **29** | Gong, et al. [91] | 3D tensor filtering by CFS method | LUNA16 dataset | N/A | N/A | N/A | 84.60% |
| **30** | Kumar, et al. [92] | CAD system with autoencoder feature | NCI Lung Image Database Consortium (LIDC) dataset | 75.01% | N/A | N/A | 83.35% |
| **31** | Wang, et al. [79] | transUnet | LIDI-IDRI database | 84.62% | N/A | 93.17% | 70.92% |
| **32** | Song, et al. [78] | Stacked autoencoder (SAE). | (LIDC) dataset | 82.59% | N/A | 81.35% | 83.96% |
| **33** | Song, et al. [78] | DNN | (LIDC) dataset | 82.37% | N/A | 83.90% | 80.66% |
| **34** | Kuruvilla and Gunavathi [93] | Neural Networks | LIDC (110) | 93.30% | N/A | 100.00% | 91.40% |
| **35** | Song, et al. [78] | Curvelet transform and NN | Private (120) | 90.00% | N/A | 93.33% | 86.66% |
| **36** | Mukherjee, et al. [94] | Ensemble stacking | LIDC-IDRI database | 99.50% | N/A | 98.80% | 99.20% |
| **37** | Wang, et al. [95] | Adaptive-boost DL | Private (1478 patients) | 73.40% | 83.80% | 76.20% | 70.50% |
| **38** | Silva, et al. [84] | ConvNet | LUNGx | 90.40% | N/A | 92.40% | 88.70% |

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| **39** | Zhang, et al. [96] | Dense-Inception U-net (DIU-net) | Lung dataset (Kaggle ) | 99.4% | N/A | 99.80% | 98.50% |
| **40** | Faruqui, et al. [97] | LungNet | LIDC-IDRCs | 96.81% | N/A | 96.65% | 97.02% |

The case studies below are classification tasks for models used to classify patients' categories or predict disease presence in Lung Cancer.

Both models from Al-Sheikh, et al. [69] show excellent performance with 98.2% accuracy on 904 X-ray images, but AlexNet has a slight edge in precision. This demonstrates the robustness of these pre-trained convolutional neural networks in handling medical image classification. SVM has been explored in various studies, showing consistent performance across datasets. For instance, Pathan, et al. [70] achieved a 96.00% accuracy on a dataset of 1000 images with a sensitivity of 100.00%, while Rodrigues, et al. [71] reached 96.70% accuracy on the LIDC-IDRI dataset. In contrast, Orozco, et al. [72] model had only 52.17% specificity, showing that dataset size and quality impact performance.

Pathan, et al. [70] KNN model reached 99.00% accuracy and 100.00% precision, outperforming many other models. Rodrigues, et al. [71] KNN implementation on the LIDC/IDRI dataset achieved 95.30% accuracy, also demonstrating that KNN remains competitive when tuned effectively. Both methods performed excellently on Rodrigues, et al. [71] dataset, with 100.00% accuracy, precision, and sensitivity, demonstrating their effectiveness in structured medical imaging datasets. However, these results are dataset-dependent, with other studies reporting lower values in real-world scenarios. CNN models have been highly effective in lung cancer detection. Masud, et al. [77] CNN on the LC25000 dataset achieved 96.33% accuracy, and Crasta, et al. [82] 3D-ResNet on the LUNA16 dataset demonstrated superior performance with 99.20% accuracy and 98.8% sensitivity. The variations in CNN results across datasets (Song, et al. [78] reported only 84.15% accuracy) highlight how architectural enhancements and dataset size influence performance.

3D CNNs, such as Liao, et al. [81] model, are designed to capture the 3D spatial relationships in medical images. It model achieved 81.40% accuracy on the LUNA16 dataset, while Crasta, et al. [82] 3D-ResNet surpassed it with 99.20% accuracy on the same dataset, indicating the significant advantage of residual networks in 3D data.Ensemble models like Mukherjee, et al. [94] stacking ensemble reached 99.50% accuracy on the LIDC-IDRI dataset, illustrating the power of combining multiple classifiers. Similarly, Sharma, et al. [75] combined SVM and KNN in a hybrid model, achieving 93.90% accuracy, showcasing the benefits of hybrid approaches in enhancing sensitivity and specificity.Wu, et al. [83] deep residual network incorporating migration learning reached 98.20% accuracy on the LIDC-IDRI dataset, emphasizing the effectiveness of deep residual architectures in medical imaging.

## Pneumonia

Tuberculosis (TB) remains a major public health issue in Africa, where it accounts for a significant number of deaths and illnesses [98, 99]. The continent experiences some of the highest TB incidence rates globally, largely due to factors such as poverty, overcrowded living conditions, and limited access to healthcare. The HIV/AIDS epidemic further exacerbates the TB burden, as co-infection weakens the immune system, making individuals more vulnerable to the disease [49, 100, 101].

*Table 4: Overview of Machine Learning Models in Pneumonia*

| Case Study | Author(s) | Model Type | Dataset Size | Accuracy | Precision | Specificity | Sensitivity |
|---|---|---|---|---|---|---|---|
| **1** | Becker, et al. [102] | Lunit INSIGHT CXR3 | 948 dyspnea patients | N/A | N/A | 66.00% | 95.40% |
| **2** | Rahman, et al. [103] | AlexNets | 5247 chest x-rays,Kaggle | 94.50% | 93.10% | 92.60% | 95.30% |
| **3** | Chouhan, et al. [104] | AlexNet | Guangzhou Women and Children's Medical Center (5232 images) | 92.86% | 90.21% | N/A | 98.97% |
| **4** | Rahman, et al. [103] | ResNet18 | 5247 chest x-rays,Kaggle | 96.40% | 95.40% | 95. 00% | 97.00% |
| **5** | Chouhan, et al. [104] | ResNet18 | Guangzhou Women and Children's Medical Center (5232 images) | 94.23% | 91.58% | N/A | 99.48% |
| **6** | Ayan, et al. [105] | ResNet-50 | Kernaby datasets. 5856 images | 94.42% | N/A | 91.45% | 95.89% |
| **7** | Mahmud, et al. [106] | ResNet | 5856 images from Guangzhou Medical center + 305 from Sylhet Medical College | 91.20% | 90.70% | 84.10% | 95.90% |
| **8** | Rahman, et al. [103] | DenseNet201 | 5247 chest x-rays, Kaggle | 98.00% | 97.00% | 97.00% | 99.00% |
| **9** | Chouhan, et al. [104] | DenseNet121 | Guangzhou Women and Children's Medical Center (5232 images) | 92.62% | 91.18% | N/A | 99.23% |
| **10** | Rajaraman, et al. [107] | Customized VGG16 | pediatric CXRs | 96.20% | 99.30% | 96.20% | 96.20% |
| **11** | Sharma and Guleria [108] | VGG16 | CXR image datasets containing 6,436 | 92.15% | 94.28% | 93.70% | 93.08% |

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|---|---|---|---|---|---|---|---|
| **12** | Ayan, et al. [105] | VCG-16 | Kernaby datasets. 5856 images | 92.94% | N/A | 90.59% | 94.35% |
| **13** | Ayan, et al. [105] | VCG-19 | Kernaby datasets. 5856 images | 91.02% | N/A | 89.31% | 90.05% |
| **14** | Mahmud, et al. [106] | VCG-19 | 5856 images from Guangzhou Medical center + 305 from Sylhet Medical College | 87.20% | 85.60% | 77.90% | 91.10% |
| **15** | Rahman, et al. [103] | SqueezeNet | 5247 chest x-rays,Kaggle | 96.10% | 98.50% | 98. 00% | 94.00% |
| **16** | Ayan, et al. [105] | SqueezeNet | Kernaby datasets. 5856 images | 94.07% | N/A | 93.88% | 94.61% |
| **17** | Kundu, et al. [109] | GoogleNet, DenseNet-121, ResNet-18 | Kernaby datasets. 5856 images | 98.81% | 98.82% | 98.79% | 98.80% |
| **18** | Kundu, et al. [109] | GoogleNet, DenseNet-121, ResNet-18 | RSNA datasets. 26601 images | 86.85% | 86.89% | 86.95% | 87.02% |
| **19** | Ayan, et al. [105] | MobileNet | Kernaby datasets. 5856 images | 94.87% | N/A | 91.45% | 96.92% |
| **20** | Trivedi and Gupta [110] | MobileNet | 5856 chest x-rays, Kaggle | 97.09% | 97.00% | 96.00% | 98. 00% |
| **21** | Ayan, et al. [105] | InceptionV3 | Kernaby datasets. 5856 images | 93.91% | N/A | 89.310% | 96.66% |
| **22** | Mahmud, et al. [106] | Inception | 5856 images from Guangzhou Medical center + 305 from Sylhet Medical College | 88.70% | 88.90% | 80.20% | 94.10% |
| **23** | Chouhan, et al. [104] | InceptionV3 | Guangzhou Women and Children's Medical Center (5232 images) | 92.01% | 90.30% | N/A | 98.46% |
| **24** | Ayan, et al. [105] | Xception | Kernaby datasets. 5856 images | 95.03% | N/A | 91.02% | 97.43% |
| **25** | Bhatt and Shah [111] | Ensemble technique | 5863 images on Guangzhou | 84.23% | 80.04% | 88.56% | 99.23% |
| **26** | An, et al. [112] | DenseNet121 and EfficientNetB0 | Guangzhou, Kaggle images. | 95.19% | 98.38% | 97.43% | 96.06% |
| **27** | Stokes, et al. [113] | Linear regression | 4500 individuals. | 83.00% | 76.00% | 88.00% | 72.00% |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | October 2017 to December 2018 | | | | |
| **28** | Stokes, et al. [113] | Decision Tree | 4500 individuals. October 2017 to December 2018 | 84.00% | 73.00% | 84.00% | 80.00% |
| **29** | Stokes, et al. [113] | Linear SVM | 4500 individuals. October 2017 to December 2018 | 82.00% | 74.00% | 85.00% | 77.00% |
| **30** | Ayan, et al. [105] | PNet | Kernaby datasets. 5856 images | 93.93% | N/A | 89.74% | 96.41% |
| **31** | Mahmud, et al. [106] | CovxNet | 5856 images from Guangzhou Medical center + 305 from Sylhet Medical College | 98.10% | 98.00% | 97.90% | 98.50% |
| **32** | Chouhan, et al. [104] | GoogleNet | Guangzhou Women and Children's Medical Center (5232 images) | 93.12% | 90.44% | N/A | 99.48% |
| **33** | Nahid, et al. [114] | Multichannel CNN | 5856 images from Guangzhou Medical center + 305 from Sylhet Medical College | 97.93% | 98.38% | N/A | 97.47% |
| **34** | Becker, et al. [115] | DL algorithm | x-ray of 948 patients with dyspnea (3 February to 8 May 2020) as well as (15 October to 15 December 2020) | 92.30% | 80.20% | 66.0 | 95.40% |
| **35** | Goyal and Singh [116] | RNN + LSTM | C19RD dataset | 94.31% | 88.89% | 90.67 | 95.41% |

The case studies below are classification tasks for models used to classify patients' categories or predict disease presence in Pneumonia.

In this comparative study of machine learning models used for chest X-ray classification, the analysis mainly revolves around models such as AlexNet, ResNet, DenseNet, VGG, and Ensemble techniques, each evaluated using diverse datasets ranging from the Guangzhou Women and Children's Medical Center to Kaggle and Kernaby datasets. The main performance indicators used are accuracy, precision, specificity, and sensitivity.

AlexNet, used in studies by Rahman, et al. [103] and Chouhan, et al. [104], shows commendable performance across different datasets. On the Kaggle dataset, Rahman, et al. [103] model attains an accuracy of 94.5%, precision of 93.10%, and a sensitivity of 95.30%, reflecting robust generalization on large-scale data. In contrast, Chouhan, et al. [104] model, though slightly lagging in accuracy at 92.86%, excels with higher sensitivity at 98.97%, indicating superior identification of true positives. Moving to ResNet variants, Rahman, et al. [103] ResNet18 on the Kaggle dataset demonstrates high overall accuracy (96.4%) and balanced precision (95.40%) and specificity (95.00%). Similarly, Chouhan, et al. [104] ResNet18 trained on the Guangzhou dataset performs slightly better, with a higher sensitivity of 99.48%. This suggests ResNet models are effective across datasets, with a particular strength in detecting conditions accurately.

The DenseNet models, specifically DenseNet121 and DenseNet201, consistently exhibit strong performance. Rahman, et al. [103] DenseNet201 on the Kaggle dataset achieves 98% accuracy, one of the highest in the study, and a near-perfect sensitivity (99.00%). Chouhan, et al. [104] DenseNet121 on the Guangzhou dataset maintains this high standard with a sensitivity of 99.23%. DenseNet models seem to outperform most other architectures in terms of sensitivity, making them highly reliable for true positive identifications. VGG16 and VGG19, while effective, show variability in their performance. For instance, Sharma and Guleria [108] VGG16 on a CXR dataset reaches an accuracy of 92.15% with moderate specificity (93.70%) and sensitivity (93.08%). Comparatively, Ayan, et al. [105] VGG19 model achieves lower accuracy (91.02%) and lower sensitivity (90.05%), which could imply a relative decline in performance with deeper versions of the VGG architecture, possibly due to overfitting or limitations in training with the given datasets.

In contrast, Ensemble techniques, such as those combining GoogleNet, DenseNet, and ResNet, appear to consistently outshine individual models. Kundu, et al. [109] ensemble model on the Kernaby dataset shows an impressive 98.81% accuracy, with all other metrics (precision, specificity, sensitivity) hovering around 98.80%, suggesting that leveraging multiple models provides a well-rounded and reliable classification. However, when applied to a larger dataset such as RSNA (with 26,601 images), the performance drops to 86.85%, highlighting the challenge of scaling ensemble methods effectively.

Lastly, models such as MobileNet and Inception yield competitive results. Ayan, et al. [105] MobileNet reaches a high accuracy of 94.87% on the Kernaby dataset, with a strong sensitivity of 96.92%, showcasing its effectiveness in lightweight and efficient model architectures. However, Mahmud, et al. [106] Inception on a smaller dataset from Guangzhou records a lower accuracy of 88.70%, suggesting that while MobileNet performs consistently across datasets, Inception might struggle in smaller-scale datasets.

## Tuberculosis

Approximately 2 million people worldwide die from tuberculosis each year[117]. The application of machine learning (ML) models in tuberculosis (TB) research and clinical practice is rapidly advancing to improve drug susceptibility testing, diagnosis, and treatment[118]. ML models have been used to effectively forecast treatment results for tuberculosis by utilizing large amounts of patient data. For example, a study's recall of 98.00% and accuracy score of 0.95 show how effective machine learning can be at predicting treatment responses[119].

*Table 5: Overview of Machine Learning Models in Tuberculosis*

| Case Study | Author(s) | Model Type | Dataset Size | Accuracy | Precision | Specificity | Sensitivity |
|---|---|---|---|---|---|---|---|
| **1** | Zhu, et al. [53] | Deep Learning | 2,983 patients | 78.20% | 88.20% | N/A | N/A |
| **2** | Amaral, et al. [54] | KNN | 4,235 patients | 97.00% | N/A | 99.00% | 96.00% |
| **3** | Maldonado-Franco, et al. [57] | DT | 695 patients | 88.60% | 84.60% | 96.20% | 64.70% |
| **4** | Amaral, et al. [54] | DT | 4,235 patients | 90.00% | N/A | 90.00% | 91.00% |
| **5** | Chinagudaba, et al. [119] | DT | 500,000 patient records | 80.80% | 87.50% | N/A | N/A |
| **6** | Amaral, et al. [54] | SVM | 4,235 patients | 96.00% | N/A | 94.00% | 97.00% |
| **7** | Orjuela-Cañón, et al. [120] | SVM | 233 patients | 81.00% | N/A | 55.00% | 95.00% |
| **8** | Ali, et al. [121] | SVM | 275 patients | 72.34% | N/A | 78.99% | 63.44% |
| **9** | Lai, et al. [122] | SVM | 127 patients | 72.67% | N/A | 78.40% | 48.00% |
| **10** | Liao, et al. [123] | SVM | 2248 TB patients | 68.60% | N/A | 68.10% | 77.80% |
| **11** | Sauer, et al. [124] | SVM | TB portals as per March 2018 | 69.00% | 56.00% | 94.00% | 21.00% |
| **12** | Chinagudaba, et al. [119] | XGBoost | 500,000 patient records | 87.90% | 87.40% | N/A | N/A |
| **13** | Zhong, et al. [125] | XGBoost | 743 patients | 76.00% | 90.00% | N/A | 74.00% |
| **14** | Liao, et al. [123] | XGBoost | 2248 TB patients | 70.50% | N/A | 70.60% | 69.40% |
| **15** | Rashidi, et al. [126] | Gradient Boosting machine | 278 subjects | 90.00% | N/A | 100.00% | 89.00% |
| **16** | Ali, et al. [121] | LR | 275 patients | 68.91% | N/A | 81.30% | 51.87% |
| **17** | Orjuela-Cañón, et al. [120] | LR | 233 patients | 86.00% | N/A | 66.00% | 94.00% |

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|---|---|---|---|---|---|---|---|
| **18** | Liao, et al. [123] | LR | 2248 TB patients | 65.60% | N/A | 65.60% | 66.70% |
| **19.** | Bajaj, et al. [127] | LR | Kaggle dataset | 85.44% | 93.12% | N/A | 93.33% |
| **20.** | Zhang, et al. [128] | Logistic Regression and lasso regression | 1,060 patients | N/A | 97.99% | 78.13% | 34.25% |
| **21.** | Sauer, et al. [124] | LASSO | TB portals as per March 2018 | 72.00% | 64.00% | 96.00% | 21.00% |
| **22** | Ali, et al. [121] | RF | 275 patients | 74.47% | N/A | 74.31% | 75.12% |
| **23** | Lai, et al. [122] | RF | 127 patients | 71.33% | N/A | 75.20% | 48.00% |
| **24** | Liao, et al. [123] | RF | 2248 TB patients | 71.30% | N/A | 71.20% | 72.20% |
| **25** | Sauer, et al. [124] | RF | TB portals as per March 2018 | 70.00% | 52.00% | 91.00% | 30.00% |
| **26** | Rashidi, et al. [126] | RF | 278 subjects | 89.00% | N/A | 100.00% | 88.00% |
| **27** | Bajaj, et al. [127] | RF | Kaggle dataset | 92.87% | 77.56% | N/A | 75.76% |
| **28** | Lai, et al. [122] | ANN | 127 patients | 88.67% | N/A | 90.40% | 80.00% |
| **29** | Bajaj, et al. [127] | ANN | Kaggle dataset | 90.34% | 91.28% | N/A | 91.26% |
| **30** | Liao, et al. [123] | MLP | 2248 TB patients | 73.50% | N/A | 73.60% | 72.20% |
| **31** | Nijiati, et al. [129] | VGG-16 | 2,291 patients | 91.20% | N/A | N/A | 89.80% |
| **32** | Nijiati, et al. [129] | EfficientNet | 2,291 patients | 94.40% | N/A | N/A | 91.00% |
| **33** | Nijiati, et al. [129] | ResNet-50 | 2,291 patients | 97.10% | N/A | N/A | 93.50% |
| **34** | Zhang, et al. [130] | DCNN | Genomatic datasets | 93.00% | N/A | N/A | N/A |
| **35** | Bajaj, et al. [127] | LSTM | Kaggle dataset | 94.90% | 95.12% | N/A | 95.22% |

The case studies below are classification tasks for models used to classify patients' categories or predict disease presence in Lung Cancer.

In examining the performance of deep learning models, Zhu, et al. [53] reported a deep learning model with an accuracy of 78.20%, achieving a strong precision of 88.20%. This model's precision highlights its ability to correctly identify positive cases but suggests room for improvement in terms of sensitivity. Similarly, Nijiati, et al. [129] employed a range of DL architectures—VGG-16, EfficientNet, and ResNet-50—with the latter two models surpassing 94.00% accuracy. Among these, ResNet-50 stood out, demonstrating a remarkable 97.10% accuracy, positioning it as one of the top performers in terms of prediction reliability. Similarly, Zhang, et al. [128] applied a deep convolutional neural network (DCNN) to genomatic datasets, reaching an accuracy of 93.00%, reflecting the potency of deep learning in complex datasets.

Support vector machines (SVM) have been popular in medical diagnostics, with diverse results across studies. Amaral, et al. [54] achieved high accuracy (96%) and strong sensitivity (97.00%) in their SVM model, highlighting its efficiency in disease prediction. However, performance varies across datasets, as Liao, et al. [123]'s work on tuberculosis (TB) patients showed a lower accuracy of 68.60%, with similarly mixed results from Sauer, et al. [124] , who reported an SVM model with 69.00% accuracy but low sensitivity (21%). Decision trees (DT) have also demonstrated potential, with Amaral, et al. [54] 's DT model achieving 90% accuracy and sensitivity (91.00%), positioning DT as a robust method for diagnostic applications. Maldonado-Franco, et al. [57] also reported a DT model with high accuracy (88.60%) and specificity (96.20%), although its sensitivity lagged behind (64.70%). In contrast, Chinagudaba, et al. [119]'s model, applied to a much larger dataset, had a lower accuracy of 80.80%, showing that larger datasets may challenge the precision of DT models.

Ensemble methods, including random forests (RF) and gradient boosting, yielded promising results across studies. Rashidi, et al. [126] applied RF and gradient boosting models to their dataset, with RF achieving an accuracy of 89.00% and gradient boosting peaking at 90.00%. These models demonstrated perfect specificity (100.00%), making them highly suitable for tasks requiring high confidence in identifying negative cases. Chinagudaba, et al. [119] leveraged the power of XGBoost, reporting an accuracy of 87.90%, while Bajaj, et al. [127] obtained even higher results with RF, reaching 92.87% accuracy, further highlighting the advantages of ensemble learning approaches in improving prediction accuracy and robustness.

Logistic regression (LR) models, while simpler than other methods, performed relatively well. Zhang, et al. [128] combined logistic regression and lasso regression, achieving near-perfect precision (97.99%), though the sensitivity was notably lower at 34.25%. Other studies using LR, such as Bajaj, et al. [127] (85.44% accuracy and 93.12% precision), suggest that despite being less complex, LR models still offer reliable prediction, particularly when precision is a priority. Other techniques, such as artificial neural networks (ANN) and multi-layer perceptrons (MLP), have also demonstrated utility. Bajaj, et al. [127] reported a strong performance from their ANN model with 90.34% accuracy and over 91.00% precision, whereas Lai, et al. [122] found that ANN could achieve an accuracy of 88.67%, coupled with high specificity (90.40%). Liao, et al. [123] 's MLP model on TB patients showed lower performance, with a 73.50% accuracy, suggesting that more complex architectures or tuning might be necessary for specific medical conditions.

# Discussion

The use of supervised learning models in pulmonary medicine has led to notable improvements in diagnostic and predictive capabilities[31, 131, 132]. This review examines various supervised learning techniques, including traditional algorithms like logistic regression and decision trees, as well as more advanced models such as random forests and support vector machines, assessing their effectiveness across a spectrum of pulmonary diseases. The analysis focuses on their ability to accurately identify and forecast outcomes for conditions like chronic obstructive pulmonary disease (COPD), asthma, pneumonia, tuberculosis, and lung cancer. The following sections summarize key findings from the research, highlighting both the achievements and challenges faced when integrating supervised learning into pulmonary diagnostics and treatment.

Several recurring themes arise from this investigation. Supervised learning models show considerable potential for enhancing the accuracy of diagnosing and predicting pulmonary diseases. Studies have found that support vector machines and random forests achieve high accuracy in forecasting COPD exacerbations, while deep learning models are promising for classifying lung cancer from imaging data[54, 133]. These developments underscore the ability of supervised learning to improve pulmonary healthcare by uncovering complex patterns in clinical data that conventional diagnostic methods might miss.

Despite these encouraging results, challenges remain. Issues such as model interpretability, performance variability across different datasets, and the necessity for high-quality, comprehensive data pose significant obstacles to the broad clinical adoption of supervised learning[134, 135]. Concerns about the generalizability of these models to diverse populations and healthcare systems persist, and ensuring their practical application in real-world settings is essential. Furthermore, the complexity of certain supervised learning methods, particularly deep learning models, can complicate interpretation for healthcare providers, raising important questions about the transparency and comprehensibility of these systems[136, 137].

The following discussion will delve into these issues, offering a balanced perspective on the potential benefits and limitations of employing supervised learning in pulmonary medicine. It will emphasize the need for interdisciplinary collaboration between healthcare practitioners and data scientists to address current challenges and identify future research avenues for refining supervised learning applications in pulmonary care. Ultimately, this review underscores that while supervised learning holds great potential, its effective integration into clinical practice will necessitate ongoing efforts to improve model reliability, transparency, and accessibility.

## Objective 1

The integration of artificial intelligence (AI) in medical diagnostics has garnered increasing attention in recent years, particularly concerning lung pathologies such as asthma, chronic obstructive pulmonary disease (COPD), pneumonia, tuberculosis, and lung cancer[19, 31, 119]. This investigation aims to synthesize findings on the effectiveness of various AI models in diagnosing these conditions, highlighting their performance metrics, advantages, and challenges. The results indicate that while AI models significantly enhance diagnostic accuracy and patient outcomes, variations in performance across different lung pathologies underscore the necessity for ongoing refinement of these technologies.

Asthma, characterized by airway inflammation and hyperreactivity, presents unique challenges for accurate diagnosis[26]. Machine learning models, particularly XGBoost and Support Vector Machines (SVM), have shown promise in this domain. For example, Luo, et al. [33] reported an XGBoost model achieving an impressive accuracy of 90.31%, albeit with a low precision of 22.65%, indicating a tendency for false positives. In contrast, Joo, et al. [34] achieved high specificity (97.90%) and sensitivity (82.50%) with the same model, underscoring how dataset quality and model tuning can drastically influence outcomes. This variability in performance highlights the complexity of asthma diagnosis, where the diversity of patient presentations and environmental factors can significantly affect model efficacy. In the case of Chronic Obstructive Pulmonary Disease (COPD), the application of AI models has demonstrated considerable promise. Convolutional Neural Networks (CNN) emerged as a particularly effective approach, with Zhu et al. reporting an accuracy of 78.20%. Moreover, Amaral, et al. [54] illustrated that K-Nearest Neighbors (KNN) could achieve accuracy rates of 97.00% while maintaining high specificity and sensitivity. These findings suggest that while deep learning models may excel in complex scenarios, simpler models like KNN can also yield remarkable results when trained on robust datasets. The variability in outcomes between studies indicates that the quality of the input data is crucial for model success in diagnosing COPD, a multifactorial condition with diverse phenotypic expressions.

Pneumonia diagnosis has seen transformative advances through AI applications, particularly deep learning architectures such as AlexNet and ResNet. Rahman, et al. [103] reported an accuracy of 94.5% with AlexNet, while ResNet models achieved an overall accuracy of 96.4% across various datasets. These results affirm the capability of deep learning to handle the complexities of pneumonia diagnostics effectively. However, despite these high accuracy rates, concerns regarding sensitivity and specificity persist, illustrating the need for careful model evaluation and consideration of clinical implications. The use of AI in pneumonia diagnosis highlights the potential for these technologies to enhance traditional diagnostic approaches, but underscores the importance of continued development and validation against clinical benchmarks. AI has also played a pivotal role in tuberculosis (TB) diagnosis, leveraging imaging techniques to detect the disease[119]. The use of machine learning algorithms in analyzing chest X-rays has shown promising results. For instance, studies utilizing CNNs have reported accuracies exceeding 90%. However, despite these advances, the challenge remains in achieving consistent sensitivity and specificity across diverse populations and imaging conditions. The global burden of TB necessitates robust AI models that can accurately identify the disease while minimizing false negatives, as this can have dire consequences for patient health and public health efforts[117].

Lung cancer diagnosis has perhaps benefited the most from AI applications, particularly with the advent of advanced deep learning techniques. Al-Sheikh et al. reported an accuracy of 98.2% using pre-trained CNNs, while Pathan, et al. [70] achieved a remarkable accuracy of 99.00% with KNN. Such high performance indicates the capability of AI to identify intricate patterns in imaging data that may elude human clinicians. Moreover, ensemble methods, such as stacking ensembles demonstrated by Mukherjee, et al. [94], reached an accuracy of 99.50%. However, while these results are encouraging, variations in sensitivity and specificity across studies highlight the need for thorough validation against clinical outcomes to ensure that AI models can reliably aid in lung cancer diagnosis. Comparing the effectiveness of AI models across different lung pathologies reveals both strengths and weaknesses inherent to each condition. For asthma and COPD, the variability in model performance suggests that while certain algorithms like XGBoost and CNNs show promise, there remains a pressing need for data quality and diversity to refine predictive capabilities[30]. The application of AI in pneumonia and TB diagnosis highlights the growing potential for deep learning to improve diagnostic accuracy; however, these models must continue to evolve to address challenges related to sensitivity and specificity. In contrast, lung cancer diagnosis stands out with the highest accuracy rates among the conditions studied, demonstrating the substantial advancements AI has brought to this area.

AI models present transformative potential for diagnosing various lung pathologies, including asthma, COPD, pneumonia, tuberculosis, and lung cancer. While the results indicate significant improvements in diagnostic accuracy, the variability in model performance across different conditions underscores the necessity for ongoing research and development. Future efforts should focus on enhancing dataset quality, optimizing model architectures, and validating these technologies against clinical standards to ensure their effective integration into real-world healthcare applications. Ultimately, the goal remains clear: to leverage AI's capabilities to improve diagnostic precision, patient outcomes, and the overall efficiency of healthcare systems in managing lung diseases.

## Objective 2

The evaluation of predictive capabilities for various machine learning models in forecasting clinical outcomes and treatment responses highlights their potential and limitations across different respiratory conditions, including asthma, COPD, lung cancer, and pneumonia.

In the case of asthma, models like XGBoost, Random Forest (RF), and Support Vector Machines (SVM) demonstrated varying degrees of success in classification tasks. For instance, Luo, et al. [33] achieved high accuracy with XGBoost (90.31%) but suffered from low precision (22.65%), suggesting that while the model was effective in identifying cases, it struggled with the accuracy of its positive predictions. Conversely, Joo, et al. [34] reported an XGBoost accuracy of 87.10%, with high specificity (97.90%) and sensitivity (82.50%), indicating that this model successfully identified most non-cases while also performing reasonably well in true positive predictions. This balance is crucial in clinical settings, where the costs of false negatives can be high. Support Vector Machines showed promising results, with Ullah, et al. [38] achieving 94.10% accuracy, high precision (94.20%), and sensitivity (98.00%), making it a robust candidate for clinical applications. However, models like the Decision Tree exhibited significant discrepancies in performance across studies, with some achieving 100% sensitivity but lower specificity. This inconsistency emphasizes the necessity of evaluating models against diverse datasets to ensure reliable clinical application. For COPD, models such as K-Nearest Neighbors (KNN), Convolutional Neural Networks (CNN), and RF illustrated the variability in predictive performance depending on the datasets used. For example, Amaral, et al. [54] recorded a KNN accuracy of 97.00% with high specificity and sensitivity, establishing its robustness in certain scenarios. However, contrasting results from Hussain et al. (86.36% accuracy) highlighted KNN's sensitivity to dataset characteristics. Similarly, SVMs showed high performance in some studies, with Amaral, et al. [54] achieving an accuracy of 96.00%, yet other studies reported significant variability in results, showcasing the model's reliance on dataset size and quality. The mixed performance of XGBoost also suggests that while boosting techniques generally enhance predictive capabilities, they may still encounter limitations in precision and sensitivity that must be carefully monitored in clinical applications.

The predictive capabilities of machine learning models in lung cancer detection were exemplified by the performance of CNNs and KNNs. Al-Sheikh, et al. [69] demonstrated the effectiveness of CNNs with an accuracy of 98.2% on X-ray images, showcasing the power of deep learning architectures in medical imaging tasks. Furthermore, KNN achieved remarkable accuracy and precision metrics, indicating its competitive nature when properly tuned[54]. However, as with other conditions, performance varied across studies, stressing the importance of dataset quality and size in determining predictive reliability. Ensemble models, such as the stacking ensemble by Mukherjee et al., achieved an impressive accuracy of 99.50%, highlighting the advantages of combining multiple classifiers to enhance sensitivity and specificity. This finding underscores the potential for ensemble approaches in improving the robustness of predictive models in clinical settings. In pneumonia classification tasks, advanced models like ResNet and DenseNet consistently exhibited high performance across multiple datasets. For instance, ResNet18 achieved a 96.4% accuracy on the Kaggle dataset, demonstrating effective generalization on large-scale data. The high sensitivity rates reported for these models further suggest their utility in identifying true positives, which is critical in clinical scenarios where timely intervention is essential.

The results from these various case studies underscore the diverse strengths and weaknesses of the predictive models examined. While many models demonstrated high accuracy, issues surrounding precision, sensitivity, and specificity present challenges in clinical adoption. The variability observed across different studies highlights the necessity of conducting further research to refine these models, optimize their parameters, and evaluate their performance on larger, more representative datasets. Ultimately, the integration of robust machine learning models into clinical practice has the potential to enhance decision-making processes, leading to improved patient outcomes and tailored treatment strategies.

## Objective 3

The integration of machine learning models into clinical practice for lung disease management presents numerous challenges, as evidenced by the performance metrics observed in various studies. This discussion analyzes these challenges, focusing on the specific objective of understanding the obstacles faced when employing machine learning models for classifying lung diseases, including asthma, COPD, lung cancer, and pneumonia.

One of the primary challenges noted is the significant variability in model performance across different studies and datasets. For instance, while some models like Support Vector Machines (SVM) demonstrated high accuracies of up to 99.73% in certain contexts, others achieved notably lower scores, indicating that the model's effectiveness can be heavily influenced by the dataset's characteristics, size, and quality. This inconsistency is further compounded by the fact that high accuracy does not always correlate with high precision or sensitivity as seen in the studies by Ullah, et al. [38], Tong, et al. [36], Topaloglu, et al. [42] and Finkelstein and Jeong [31] . Also, an XGBoost study by Muro, et al. [60] achieved an accuracy of 91.70%, but with a precision as low as 10.45% in one instance. This discrepancy highlights the difficulty in ensuring reliable model performance across varying patient populations and clinical scenarios, a key consideration when integrating these models into routine clinical practice.

A critical aspect of lung disease management is the accurate identification of patients who require treatment. Many studies, such as those by Ullah, et al. [38] and Xie and Xu [35], demonstrated models with impressive accuracies but low precision and sensitivity. For example, XGBoost's performance ranged from 87.10% accuracy with high specificity but low sensitivity of 14.47%. This poses a significant challenge in clinical settings where failing to accurately identify disease presence can lead to misdiagnoses and inadequate treatment, underscoring the need for models that balance accuracy with reliable sensitivity and precision.

The observed performance of machine learning models highlights a prevalent issue: dataset dependency. The results from studies utilizing diverse datasets revealed that performance metrics vary widely based on dataset composition, size, and inherent biases. For example, Masud, et al. [77]'s Convolutional Neural Networks (CNN) model achieved a commendable accuracy of 96.33% in one study while performing poorly in another, reflecting how critical dataset characteristics can influence model efficacy. This inconsistency raises concerns about the generalizability of these models in real-world settings, where datasets may differ significantly from those used in training and validation phases.

Finally, ethical considerations and patient safety must be paramount when integrating machine learning models into clinical practice. The variability in performance and potential for misdiagnosis raise ethical questions regarding the use of automated decision-making tools in healthcare. Ensuring that these models adhere to rigorous ethical standards and maintain patient safety is essential, necessitating robust validation processes and ongoing monitoring of model performance in real-world settings.

# Limitations

This review of machine learning applications in pulmonary diseases—such as asthma, chronic obstructive pulmonary disease (COPD), pneumonia, tuberculosis, and lung cancer—highlights several key challenges that impede the full adoption of these technologies in clinical settings. One major issue is the fast-paced advancement of machine learning models, which can cause older models to become obsolete quickly, reducing their relevance for long-term use. Moreover, the inconsistency in performance metrics and the lack of uniform datasets across different studies make it difficult to compare the effectiveness of various machine learning models accurately. This inconsistency complicates the evaluation of models' efficacy, especially when applied to diverse pulmonary diseases[132].

Another significant limitation is publication bias, where studies reporting high-performing models are more likely to be published, resulting in a skewed view of machine learning's actual effectiveness in pulmonary medicine. Many commonly used models—such as logistic regression (LR), decision trees (DT), random forest (RF), and convolutional neural networks (CNNs)—face criticism for their lack of interpretability. For clinicians, understanding how a model arrives at its conclusions is crucial for trust and practical use in patient care.

Data quality and availability present additional obstacles. Effective machine learning models depend on large, high-quality datasets, which are often limited or inconsistent in pulmonary research. Issues such as missing data, small sample sizes, and unstructured patient records can reduce a model's performance and generalizability. Ethical concerns, including patient privacy and potential biases in algorithmic decision-making, further complicate the situation. Biases in training data could lead to unequal treatment outcomes for different patient groups, raising important questions about the fairness and equity of machine learning applications. Addressing these limitations, such as enhancing data accessibility, improving model transparency, and tackling ethical concerns, will be essential for successfully integrating machine learning into pulmonary diagnostics and treatment.

# Conclusion

The application of Artificial Intelligence (AI) and machine learning (ML) in diagnosing and predicting outcomes in lung pathologies represents a transformative leap in medical practice. This narrative review has underscored the urgent need for advanced diagnostic tools and predictive models, given the alarming global prevalence of lung diseases such as asthma, chronic obstructive pulmonary disease (COPD), pneumonia, tuberculosis, and lung cancer. Each of these conditions presents unique challenges in diagnosis and treatment, often complicated by their diverse pathophysiological features.

The investigation into AI models has revealed their significant role in enhancing diagnostic accuracy and clinical decision-making. By analyzing various data types, including imaging and auscultation signals, these models have demonstrated remarkable performance in identifying and classifying lung disorders. For example, the PulmoNet model, employing deep learning techniques, has achieved accuracy rates of up to 99.40% in diagnosing conditions like pneumonia and COVID-19 [4, 19]. Similarly, studies employing models such as Support Vector Machines (SVM) and Random Forest (RF) have shown substantial success in identifying subtypes of lung cancer, achieving accuracy values of 87.60% and 86.30%, respectively.

This review has also evaluated the predictive capabilities of AI and ML models in forecasting clinical outcomes and treatment responses. The potential to predict severe exacerbations in COPD patients or the likelihood of adverse outcomes in pneumonia cases signifies a profound advancement in patient management and care. Furthermore, integrating clinical and genomic data into these models holds promise for improving the detection of biomarkers and tailoring personalized treatment strategies.

However, the integration of machine learning models into clinical practice is not without challenges. Variability in model performance across different datasets, the need for substantial computational resources, and the necessity of rigorous validation processes are critical considerations that must be addressed. Future research must focus on overcoming these limitations to fully harness the capabilities of AI and ML in lung disease management.

Ultimately, the objectives of this review—to investigate the role of AI models in diagnosing lung pathologies, evaluate their predictive capabilities, and analyze the challenges associated with their integration into clinical practice—highlight the necessity for continued exploration and refinement of these technologies. As we move forward, fostering collaboration between clinicians, researchers, and data scientists will be essential in realizing the full potential of AI and ML, ultimately improving outcomes for patients with lung pathologies and reshaping the landscape of respiratory medicine.

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